SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

MPSTME LIBRARY NMIMS UNIVERSITY

Program/s: B Tech/MBA Tech/B Tech (Integrated) Year: II/III/IV Semester: III/V/VII

Stream/s ; IT, CE, CS, MXTC, EXTC, DS, AIDS, AIML, AI, CSE DS, CSE Cyber Sec

Subject: Data Structure and Algorithms Time: 3 hrs (10 .00 am to 1.00 pm)

Date: 14 / 02 / 2024 No. of Pages: 4

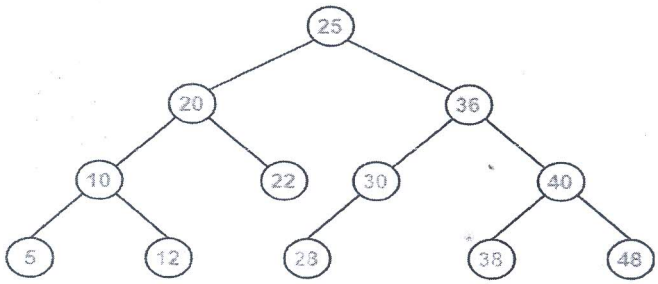
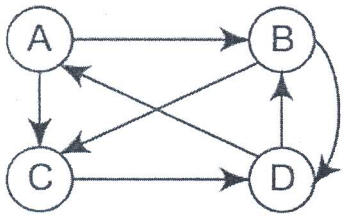
Marks: 100

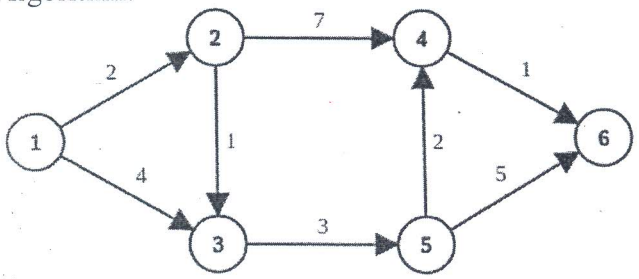
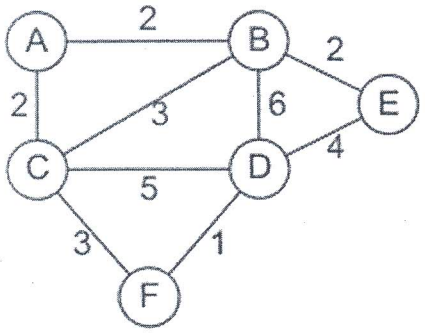
Re-Examination

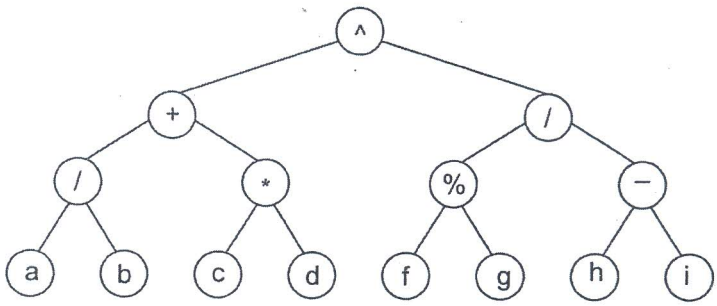
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
	a.	What is recursion? Give recursive algorithm to calculate the factorial of the given number.	[5]
CO-1,3 ; SO- 1,6; BL-1,2,3	b.	The memory representation of the marks of Chemistry is given below. Empty places shows that these students didn't appear the exam. Roll Number / Marks 1 / 50 2 / 3 / 60 4 / 20 5 / 30 6 / 7 / 40 8 / 9 / 90 10 / 35 I. Draw the memory representation of linked list for storing the student's marks in ascending order. [2]	[5]

		II. Draw the memory representation of linked list if roll number 3 and 7 leaves the program in linked list created in I [2] III. Draw the memory representation of linked list after allocating Roll number 8 gives the exam later and got 59 marks. [1]	
	c.	Find the degree of Node 20, Siblings of Node 10, level of the tree, height of Node 36 from the given binary tree.  	[5]
	d.	Give the adjacency matrix and adjacency list for the below given graph.  	[5]
Q2 CO-1,2; SO-1,4; BL-2,3	(a)	i. Define time and space complexity and find the time complexity of given algorithm **Sum (A, n)** **{** **s = 0;** **for(i=0; i<n; i++)** **{** **s = s + A[i];** **}** **return s;** **}** ii. Write an algorithm to evaluate postfix expression using stack and apply the algorithm on given postfix expression to evaluate it. 9 3 4 * 8 + 4 / –	[10]
	(b)	Explain Queue as a data structure and its application using any real life example? Write an algorithm or function to insert and delete an element in queue.	[10]

Q3 CO-2,3; SO-1,6; BL-1,2,3	(a)	Explain the steps in Dijkstra's algorithm. Find the shortest path from vertex 1 to remaining all vertices for the graph given below using Dijkstra's Algorithm.  	[10]
	(b)	Define infix, postfix, and prefix expression? Write an algorithm to convert infix expression to postfix expression. Convert given infix expression in to postfix expression. **A+(B*C(D/E^F)*G)*H**	[10]
Q4 CO-2,4; SO-1; BL-2,3	(a)	What is doubly linked list? Why it is used? Write a function/ algorithm to implement stack using linked list with following cases: i. When stack is empty ii. When stack is full	[10]
	(b)	i. Construct a Binary Search Tree (BST) for the following sequence of numbers and Find the number of nodes in the left subtree and right subtree of the root. 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24 ii. Write an algorithm of postorder traversal in binary tree. Construct the binary tree whose traversals are given below. Inorder: {4, 8, 2, 5, 1, 6, 3, 7}, Postorder: {8, 4, 5, 2, 6, 7, 3, 1}	[10]
Q5 CO-3; SO-1; BL-1,2	(a)	Explain steps in Prim's Algorithm. Find the minimum spanning tree and the cost of the graph given below using Prim's algorithm. Consider vertex A as starting vertex.  	[10]

	(b)	Write an algorithm/Function to insert a node in binary search tree (BST). Construct a Binary search tree using the given numbers. 13, 3, 4, 12, 14, 10, 5, 1, 8, 2, 7, 9, 11, 6, 18.	[10]
Q6 CO-2,3,4; SO-1; BL- 1,2,3	(a)	What is Linked list? Write a simple function / algorithm for i) Inserting a new node after a given value in Singly Linked List ii) Deleting a node from the beginning in the Singly linked list	[10]
	(b)	Write an algorithm of merge sort with its time complexity. Sort the given element using merge sort and represent all the steps. 39, 9, 81, 45, 90, 27, 72, 18.	[10]
Q7 CO-2,3,4; SO-1; BL-1,2,3	(a)	Write the expression for the given binary tree. What is linear probing in hashing? Insert the elements 50, 70, 76, 93 using linear probing considering the mod 5.  	[5]
	(b)	What do you understand by the concept of circular queue? How is it better than a linear queue?	[5]
	(c)	Compare selection sort and insertion sort algorithms.	[5]
	(d)	Write an algorithm to explain Depth First Search with suitable example.	[5]